after this treatment are allowed to dry and then sown. The plants will show quick flowering when compared to untreated control plants.

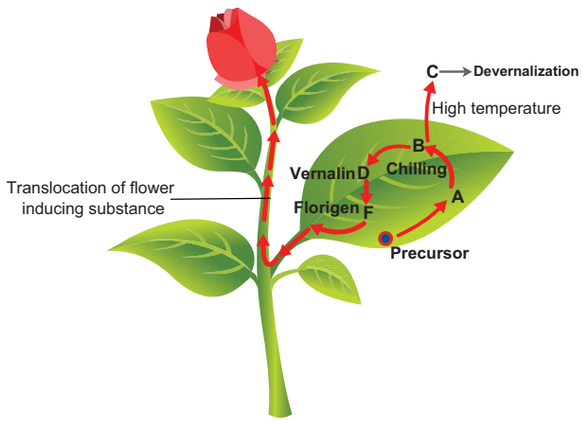


Figure 15.21: Vernalization and Flowering

3. Devernalization

Reversal of the effect of vernalization is called **devernalization**.

4. Practical applications

1. Vernalization shortens the vegetative period and induces the plant to flower earlier.
2. It increases the cold resistance of the plants.
3. It increases the resistance of plants to fungal disease.
4. Plant breeding can be accelerated.

15.6 Seed Germination and Dormancy

I. Seed Germination

The activation and growth of embryo from seed into seedling during favourable conditions is called **seed germination**.

1. Types of germination

There are two methods of seed germination. Epigeal and hypogeal.

i. Epigeal germination

During epigeal germination cotyledons are pushed out of the soil. This happens due to the elongation of the hypocotyl. Example: Castor and Bean.

ii. Hypogeal germination

During hypogeal germination cotyledons remain below the soil due to rapid elongation of epicotyls (Figure 15.22). Example: Maize, Pea.

2. Factors affecting germination

Seed germination is directly affected by external and internal factors:

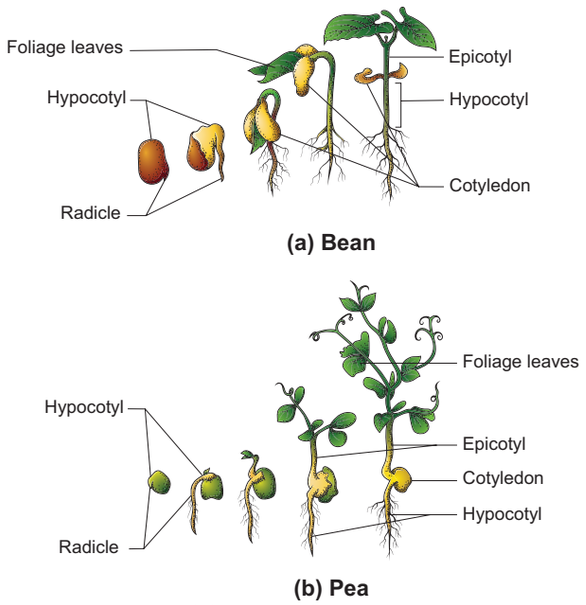


Figure 15.22: (a) Epigeal Germination
(b) Hypogeal Germination

i. External factors

a. **Water**: It activates the enzymes which digest the complex reserve foods of the seed. If the water content of the seed goes below a critical level, seeds fail to germinate.

b. **Temperature**: Seeds fails to germinate at very low and high temperature. The optimum temperature is 25°C to 35°C for most tropic species.

c. **Oxygen**: It is necessary for germination. Since aerobic respiration is a physiological requirement for germination most will germinate well in air containing 20% oxygen.

d. **Light**: There are many seeds which respond to light for germination and these seeds said to be photoblastic.

e. **Soil conditions**: Germination of seed in its natural habit is influenced by soil conditions such as water holding capacity, mineral composition and aeration of the soil.

ii. Internal factors

a. **Maturity of embryo**: The seeds of some plants, when shed will contain immature embryo. Such seeds germinate only after maturation of embryo.

b. **Viability**: Usually seeds remain viable or living only for a particular period. Viability

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